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DEPARTMENT OF DEFENSE HANDBOOK

COLD SPRAY NONDESTRUCTIVE TESTING METHODS FOR QUALITY ASSURANCE

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FORWARD

1. This handbook is approved for use by all departments and agencies of the Department of Defense (DoD).

2. This handbook is for guidance only and cannot be cited as a requirement.

3. This handbook provides DoD recommended guidance to address nondestructive testing methods and quality assurance processes in cold spray applications.

4. Comments, suggestions, or questions on this document should be addressed to: Director, U.S. Army DEVCOM, Army Research Laboratory (ARL), Weapons and Materials Research Directorate, Specifications & Standards Office, Attn: FCDD-RLA-M, Aberdeen Proving Ground, MD 21005-5069. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

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1. SCOPE

1.1 Purpose. This handbook provides Department of Defense (DoD) recommended guidance to address nondestructive testing (NDT) methods and quality assurance (QA) processes in cold spray applications. It covers common NDT methods that are applicable to Cold Spray processes including visual, liquid penetrant, magnetic particle, eddy current, ultrasound, radiography, and thermography. The effectiveness of each NDT method is discussed within this document.

1.2 Background. Cold spray additive repair is a topic championed by the Joint Defense Manufacturing Council (JDMC), a forum of 2-star leaders that provides strategic oversight and alignment to manufacturing technology related working groups and activities across the DoD. In September 2020, the JDMC established the Cold Spray Cross-Service Group, also known as the Cold Spray Working Group. The Cold Spray Working Group (1) influences and accelerates adoption of cold spray technology by addressing systemic opportunities and challenges that reach across the DoD and (2) facilitates development of cold spray related joint strategies thereby avoiding redundancy and duplication of efforts. Goals are achieved by addressing current or planned cold spray activities and efforts in seven areas-of-opportunity (or lines-of-effort). These lines-of-effort include:

1. Technology Approval Process
2. Material, Process, and Property Relationships
3. Common Database Approach and Taxonomy for Data
4. Nondestructive Testing and Quality Assurance
5. Facilitate Site Installation and Approval to Operate
6. Training and Certification Standards
7. Use-Case Analysis

The primary focus area of this handbook is cold spray NDT&QA.

1.2.1 Personnel Certification.

1.2.1.1 Air Force. The Air Force currently recognizes National Aerospace Standard (NAS) 410, Certification Qualification of Nondestructive Test Personnel, as the approved standard for qualification and certification. This standard establishes minimum training, experience, and examination requirements for personnel performing NDI in the aerospace manufacturing, service, maintenance, and overhaul industry. All Air Force Field civil service nondestructive inspection technicians SHOULD be certified and qualified IAW Appendix A (exception: Technicians who hold a full-time civil service position matched with a military position in their ANG or AFRC unit will be certified IAW Paragraph 1.2.3.1.1). Depot civil service nondestructive inspection technicians are certified to NAS410 in accordance with AFSCI 20-114, Qualification of Nondestructive Inspection Personnel. Contracted lab personnel (managed by private contractor or Civil Service) SHOULD be certified by their employer in accordance with NAS410. NAS410 provides the minimum requirements for the qualification and certification of personnel conducting nondestructive testing. Compliance of contracted lab personnel SHOULD be audited and validated by the Air Force NDI Office (AFLCMC/EZPT-NDIO), Tinker AFB, OK at any time deemed necessary by the Air Force to ensure certification of contracted NDI personnel is in compliance with the requirements of NAS410. Additionally, the Air Force NDI Office should review third parties providing certification services to contractor operated, AF-owned NDI laboratories, to ensure compliance with NAS410.

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1.2.1.2 Army. DOD and non-DOD Civilian inspectors should be certified IAW NAS410.

1.2.1.3 Navy.

1.2.1.3.1 NAVAIR. All civilian personnel performing these inspections are to be qualified and certified in accordance with NAS410. Military personnel performing these inspections are to be qualified as NDI technician MOS 6033/NEC 763B in accordance with COMNAVAIRFORINST 4790.2, Naval Aviation Maintenance Program.

1.2.1.3.2 NAVSEA. Requirements for conducting NDT, which are used in determining the presence of surface or internal flaws in metals, are contained in Naval Sea Systems Command Technical Publication T9074-AS-GIB-010/271. This publication also covers minimum requirements necessary to qualify NDT personnel, procedures, and equipment.

1.3 Limitations. This handbook is for guidance only and cannot be cited as a requirement. The acceptance criteria for NDT methods in cold spray have not been fully established and will need better definition as cold spray expands as an accepted repair practice.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed below are not necessarily all of the documents referenced herein but are those needed to understand the information provided by this handbook.

2.2 Government Documents.

2.2.1 Specifications, Standards, and Handbooks. The following specifications, standards, and handbooks form part of this document to the extent specified.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-32495	-	Powders for Cold Spray Deposition

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1530	-	Aircraft Structural Integrity Program
MIL-STD-2035	-	Department of Defense Test Method: Nondestructive Testing Acceptance Criteria
MIL-STD-3021	-	Materials Deposition, Cold Spray

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.2.2 Other Government Documents, Drawings, and Publications.

DOD TECHNICAL MANUALS

T.O. 33B-1-1 NAVAIR 01-1A-16-1 TM 1-1500-335-23	-	Nondestructive Inspection Methods, Basic Theory

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T.O. 33B-1-2 NAVAIR 01-1A-16-2 TM 1-1500-366-23	-	Nondestructive Inspection, General Procedures and Process Controls

(Copies of these documents are available online at https://www.tinker.af.mil/.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

ST9074-AS-GIB-010/271	-	Requirements for Nondestructive Testing Methods
T00011542-GIB-010	-	Cold Spray Processes for Non-Structural Naval Ship Applications

(Copies of this document are available online via Model Based Product Support (MBPS) at https://mbps.navseaplm.navy.mil/Windchill/app/.)

UNITED STATES AIR FORCE SUSTAINMENT CENTER INSTRUCTIONS

AFSCI 20-114	-	Qualification of Nondestructive Inspection Personnel

(Copies of this document are available online at https://www.e-publishing.af.mil/.)

UNITED STATES AIR FORCE AIRWORTHINESS BULLETINS

AWB 100A	-	Airworthiness Process Overview and Terminology
AWB 110	-	Airworthiness Planning
AWB 120A	-	Modification Airworthiness, Relatedness and Reportability Determination
AWB 130	-	Certification Basis
AWB 140	-	Compliance Report
AWB 160A	-	Airworthiness Approvals
AWB 225	-	Airworthiness Delegated Technical Authorities
EN-ASTB-14-001	-	Cold Spray as a Repair Technology
EZ-SB-13-001	-	Material Substitution

(Copies of these documents are available online at:

https://usaf.dps.mil/teams/23230/Airworthiness/SitePages/Bulletins.aspx.)

2.3 Non-Government Publications. The following documents form a part of this document to the extent specified herein.

NATIONAL AEROSPACE STANDARDS

NAS410	-	NAS Certification & Qualification of Nondestructive Test Personnel

(Copies of this document are available online at https://www.aia-aerospace.org.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS STANDARDS

ASTM E587	-	Standard Practice for Ultrasonic Angle-Beam Contact Testing
ASTM E709	-	Standard Guide for Magnetic Particle Testing
ASTM E797	-	Standard Practice for Measuring Thickness by Manual Ultrasonic Pulse-Echo Contact Method
ASTM E1001	-	Standard Practice for Detection and Evaluation of Discontinuities by the Immersed Pulse-Echo Ultrasonic Method Using Longitudinal Waves
ASTM E1004	-	Standard Test Method for Determining Electrical Conductivity Using the Electromagnetic (Eddy Current) Method
ASTM E1417/E1417M	-	Standard Practice for Liquid Penetrant Testing
ASTM E1441	-	Standard Guide for Computed Tomography (CT) Imaging
ASTM E1444/E1444M	-	Standard Practice for Magnetic Particle Testing for Aerospace
ASTM E1742/E1742M	-	Standard Guide for Radiographic Examination
ASTM E2033/E2033M	-	Standard Practice for Computed Radiology (Photostimulable Luminescence Method)
ASTM E2445/E2445M	-	Standard Practice for Performing Evaluation and Long-Term Stability of Computed Radiography Systems
ASTM E2582	-	Standard Practice for Infrared Flash Thermography of Composite Panels and Repair Patches Used in Aerospace Applications
ASTM E2698	-	Standard Practice for Radiological Examination Using Digital Detector Arrays
ASTM E2736	-	Standard Guide for Digital Detector Array Radiology
ASTM E2737	-	Standard Practice for Digital Detector Array Performance Evaluation and Long-Term Stability
ASTM E3375	-	Standard Practice for Cone Beam Computed Tomographic (CT) Examination

(Copies of these documents are available online at https://www.astm.org/.)

SAE INTERNATIONAL STANDARDS / Aerospace Material Specifications

AS6870	-	Nondestructive Inspection Program Requirements for Aerospace Systems

(Copies of these documents are available from https://www.sae.org).

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2.4 Order of Precedence. Nothing in this document supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. DEFINITIONS AND ACRONYMS

3.1 Definitions.

Air system. An air vehicle plus the training and support systems for the air vehicle (e.g., communications, control, ground/surface/control station, launch and recovery, and support elements), and any weapons to be employed on the air vehicle.

Airworthiness (AW). The property of an air system configuration to safely attain, sustain, and complete flight in accordance with approved usage limits.

Chief Engineer. The senior (having precedence in making decisions) responsible engineer in a program office.

Cognizant Engineering Authority (CEA). The member of the procuring activity that provides the technical input for the contract or purchase agreement.

Cold Spray (CS). A material addition process that deposits layers of metal and non-metal materials on a surface under computer or manual control.

Defect. One or more flaws whose aggregate size, shape, orientation, location, or properties do not meet specified acceptance criteria and are rejectable.

Director of Engineering (DOE). Is accountable to the program executive officer for oversight of the portfolio's engineering functional support.

Discontinuity. A lack of continuity or cohesion; an intentional or unintentional interruption in the physical structure or configuration of a material or component.

Ferrous. Containing, including, or relating to iron.

Ferromagnetic. Relating to substances with a high magnetic permeability.

Flaw. An imperfection or discontinuity that may be detectable by nondestructive testing and is not necessarily rejectable.

Indication. The response or evidence from a nondestructive examination.

Metal. Any of various opaque, fusible, ductile, and typically lustrous substances that are good conductors of electricity and heat, form cations by loss of electrons, and yield basic oxides and hydroxides.

Nondestructive Testing (NDT). Method used to evaluate the material properties of a manufactured component without damaging or destroying the part.

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Non-ferrous. Not containing, including, or relating to iron.

Non-ferromagnetic. Not relating to substances with a high magnetic permeability.

Non-metal. An element or substance that is not a metal.

Non-structural Applications. For applications where the cold spray deposit is not necessary for the component to meet the load-bearing requirements and where failure or disbonds do not cause significant damage or reduction in service life.

Quality Assurance (QA). A program for the systematic monitoring and evaluation of the various aspects of a project, service, or facility to ensure that standards of quality are being met.

Structural Applications. For applications where the cold spray deposit is a significant load-bearing member or where failure or disbonds could cause significant damage or reduction in service life.

3.2 Acronyms.

ASTM	American Society for Testing and Materials
AWB	Air Worthiness Bulletin
CE	Chief Engineer
CEA	Cognizant Engineering Authority
DoD	Department of Defense
DOE	Director of Engineering
DTA	Delegated Technical Authority
JDMC	Joint Defense Manufacturing Council
MIL-DTL	Military Detail
MIL-STD	Military Standard
NDT	Nondestructive Testing
PPE	Personal Protective Equipment
QA	Quality Assurance
UIPI	Uniform Industrial Process Instruction
USAF	United States Air Force
UVA	Ultraviolet A

4. GENERAL GUIDANCE

4.1 General. Spray coating is a process where metallic and non-metallic particles are deposited to form a coating on a substrate. Common spray coating processes include thermal spray and cold spray, where thermal spray deposits material in a molten or semi-molten state and cold spray deposits material in a solid state. Cold spray is sometimes referred to as a subset of thermal spray. The hardness and adhesion of sprayed deposits repaired using cold spray are an essential part of QA that NDT methods can be used to indirectly assess.

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4.2 Cold Spray. Cold spray is a material addition process that deposits layers of powdered metal and non-metal materials on a surface under computer or manual control. During this process, a high-speed gas jet accelerates powdered particles toward a substrate. The powder particles then plastically deform and adhere to the substrate upon impact, creating a solid-state bond. These particles (1) form a coating, (2) provide dimensional restoration, and/or (3) produce bulk material or near net shapes. For near net shapes, the process typically deposits material and then the part is machined to final dimensions. It is often utilized in situations where conventional thermal spray technology cannot be successfully used. See cold spray process illustration in Appendix A.

4.3 Quality Assurance (QA). QA is a monitoring approach to evaluate various aspects of manufacturing, repair, and testing processes. It is focused on providing confidence that quality requirements will be met and to ensure consistency and repeatability. QA includes regulation of the quality of raw materials, assemblies, products and components, services related to production, and management, production, and inspection processes. QA is vital to all manufacturing, repair, and testing to ensure consistency and repeatability and that the component is free of defects. Defects include, but are not limited to, discontinuities, flaws, porosity, cracks, and inclusions.

4.4 Nondestructive Testing. NDT (also referred to as nondestructive evaluation, nondestructive examination, or nondestructive inspection) is crucial to manufacturing to verify the level of product quality. NDT consists of methods used to inspect the surface and internal structure of materials to evaluate the material properties of the component without damaging or destroying the part. This is in contrast with methods that are destructive in nature, which are done on a limited number of samples. See Appendix B for additional guidance on NDT material type. NDT can be excellent methods of assessing quality.

4.5 Nondestructive Testing Methods. The choice of which NDT method to use is based on a variety of different factors including fabrication method, material specifications, geometry or other referencing documents in which the acceptance/rejection criteria are specified. The common NDT methods include visual, liquid penetrant, magnetic particle, eddy current, ultrasonic, and radiographic testing. Keep in mind that there are limitations or strengths to each method when applied to certain materials and geometries.

4.5.1 Visual Testing. Visual testing is a method for detecting surface flaws that are directly visible to the naked eye and/ or with the help of optical devices, such as: magnification, borescopes, cameras, optical scanners, micrometers, and other devices.

A. ADVANTAGES

VT requires minimal equipment/ training. It is an expedient, versatile, and typically low-cost method to execute.

B. DISADVANTAGES

VT requires direct line of sight access to the surface of inspection (surface only technique). Interpretation of the applied cold spray results using VT is subjective by nature and the data collected may vary significantly based upon the assigned observer.

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C. APPROACHES FOR COLD SPRAY APPLICATIONS

Pre-Cold Spray: VT can be used as a preliminary check on the substrate before a cold spray application to detect loose material, cracks, and/ or other noncompliant conditions.

Post-Cold Spray: VT can be used to detect gross surface defects (overspray/ under spray areas), large gaps in the spray, and other visual indications.

4.5.2 Liquid Penetrant Testing (PT). Liquid penetrant testing is a method for revealing surface breaking flaws by bleed-out of a colored or fluorescent dye from the flaw. Discontinuities that intersect the surface can be detected by using liquid penetrants. The method may be applied to metals and non-metals. This inspection method involves applying a liquid with a contrasting dye to the test part. During a dwell period, the liquid penetrates into surface defects through capillary action. Excessive penetrant is then removed off the surface and the remaining trapped penetrant is then drawn to the surface using a developer to reveal any flaws. ASTM E1417 provides additional details in performing this method.

A. ADVANTAGES

PT can detect small indications open to the surface. Interpretation of indications is relatively straightforward. PT inspection equipment is generally simple, and operators can typically perform this inspection in the field.

B. DISADVANTAGES

PT requires the use of Personal Protective Equipment (PPE) to protect eyes and skin from chemicals, eyes from UVA radiation, etc. For fluorescent PT inspections, these inspections must be executed in darkened areas. This method utilizes chemicals that can attack nonmetallic materials (rubber/ plastics) and is not ideal for porous materials because it produces a high level of background interference that can make it difficult to interpret the results. PT only detects indications/flaws that are on or open to the surface and is unable to detect obstructed subsurface features. False positive interpretations from PT are possible due to the inherent interfaces including substrate gradients or repaired corrosion pits produced as a result of cold spray that could resemble cracks or voids.

C. APPROACHES FOR COLD SPRAY APPLICATIONS

Pre-Cold Spray: In the case of repairs, PT can be used to assess whether the surface has been adequately prepared prior to application of the cold spray material. PT could be utilized to inspect the surface of the substrate for cracks, pores, remaining pits from corrosion, and other flaws that could be open to the surface.

Post-Cold Spray: PT could also be utilized to inspect the surface of the deposit post-spray for cracks, pores, and other flaws that could be open to the surface. Depending on the sensitivity of penetrant selected, this method may only be appropriate after machining the repair. As-sprayed surface roughness may produce significant background interference, making it difficult to interpret results.

4.5.3 Magnetic Particle Testing (MT). Magnetic particle testing is a method for detecting surface and subsurface flaws by establishing a magnetic field in the metal and using iron filings to see if the field lines are constant. When this method is applied to ferromagnetic metals it has the capability to detect surface/near-surface flaws. In ferromagnetic substrates, very small ferrous

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particles are attracted to magnetic field leakage sites, just as iron filings are attracted to the poles of a magnet. Any flaws at or near the surface cause disruptions in the magnetic field of magnetized parts, in turn attract the ferrous particles to the leakage site. This allows the inspector to visualize where the flaws are located within the part. With very limited exceptions, it is not applicable to non-ferromagnetic metals and non-metals (e.g., aluminum, magnesium, titanium, and composite materials), which are typically used in aircraft construction. ASTM E709 and E1444 provide additional details in performing this method.

A. ADVANTAGES

MT requires less surface preparation to be able to detect subsurface flaws. MT is able to inspect through some thin coatings.

B. DISADVANTAGES

MT requires the use of PPE to protect eyes and skin from chemicals, eyes from UVA radiation, etc. MT is usually not applicable to non-ferromagnetic metals and non-metals e.g., aluminum, magnesium, titanium, and composite materials. For fluorescent MT inspections, these inspections must be executed in darkened areas. After the inspection is complete, a demagnetization might be required. MT is only for use with ferromagnetic materials and is not sensitive to small, rounded indications.

C. APPROACHES FOR COLD SPRAY APPLICATIONS

Pre-Cold Spray: MT may be used to inspect the surface/near-surface of ferromagnetic substrates for cracks, and other flaws.

Post-Cold Spray: MT may also be utilized to inspect the surface/near-surface of the ferromagnetic deposit for defects and other features of interest.

4.5.4 Eddy Current Testing (ET). Eddy current testing is a method used on conductive materials that has the capability to detect surface/near-surface flaws. An eddy current is a circulating electrical current induced in a conductor by an alternating magnetic field. The method is based on the interaction between a magnetic field source and the metal surface, which induces eddy currents. When eddy currents encounter a discontinuity or other obstacle, the normal path and strength of the currents is changed and can be detected. The magnitude of the response is dependent on the eddy current's frequency and amplitude, magnetic permeability, metal electrical conductivity, and the size, shape, orientation of the flaw. ASTM E1004 provides additional details in performing this method.

A. ADVANTAGES

ET is sensitive to surface and near-surface discontinuities. It can inspect through nonconductive coatings up to a certain coating thickness dependent on the conductive to nonconductive coating thickness ratio.

B. DISADVANTAGES

This method is only applicable to conductive substrates. ET requires a probe to be in contact with the part being inspected. Probe liftoff is a concern for complex, rough, and/or uneven geometries and small probe size can lead to long inspection times for large areas. ET inspection requires the use of a material specific reference standard that provides a uniform

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conductivity and is representative of the component to be inspected. It can be difficult to create an appropriate reference standard for a cold spray process since the spray process changes the conductivity of the localized material. This change in conductivity can be dependent on many different variables such as the thickness and quality of the cold spray layer. To date, uniform and consistent reference standards for cold spray have not been successfully produced so ET is currently not viable as a standalone method.

C. APPROACHES FOR COLD SPRAY APPLICATIONS

Pre-Cold Spray: In the case of repairs, ET can be used on conductive materials to assess whether the surface has been adequately prepared prior to application of the cold spray material. ET could be utilized to inspect the surface/near-surface of the substrate for cracks, pores, remaining pits from corrosion, and other flaws that could be open or close to the surface.

Post-Cold Spray: ET may also be utilized to inspect for flaws at the surface/near-surface of metallic/conductive materials if a material specific reference standard can be fabricated with a uniform conductivity that is representative of the component to be inspected.

4.5.5 Ultrasonic Testing (UT). Ultrasonic testing is a method that interrogates materials using the transmission of high-frequency sound waves to travel through the component to detect indications. This method typically uses a piezoelectric ultrasound transducer to characterize material thickness or internal structure. When the sound traveling through the material encounters a discontinuity, it alters the path of travel of the sound and therefore can be detected. It is widely used on metals and non-metals including plastics, composites, and ceramics. ASTM E587, E797, and E1001 provide additional details in performing this method.

A. ADVANTAGES

UT is useful for nearly all substrates and can detect flaws from the surface to deep within the material to be inspected. Additionally, the inspection can yield significant information about flaws to include: size, orientation, and location within the component.

B. DISADVANTAGES

Since UT requires a coupling process between the transducer and the test material, physical contact via the transducer may be limited in certain situations due to geometries of the component to be inspected. Custom transducers could be developed to mitigate this disadvantage but could impact schedule, cost, and interpretation of results. Advanced ultrasonic techniques are needed to resolve multilayered indications or features of interest inherent to cold spray deposits. Some material systems may rapidly attenuate ultrasound signals, which can make inspection difficult or impossible. Complex geometrical features can make interpretation difficult or decrease inspectability.

C. APPROACHES FOR COLD SPRAY APPLICATIONS

Pre-Cold Spray: UT could be used to inspect for flaws at the surface/near-surface and deep within the component. It can also be used to determine remaining substrate thickness prior to cold spray application.

Post-Cold Spray: UT could also be utilized to inspect for flaws at the surface/near-surface and throughout the volume of the cold spray and component.

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4.5.6 Radiography. Radiographic Testing (RT) is a method that uses either X-rays or gamma rays (i.e., cesium-137, cobalt-60, and iridium-192) and film/detectors to examine the internal structure of a material to identify flaws. This method is based on detecting changes in the material of interest via alterations to the amount of exposure (radiation energy) that reaches a particular area of the detector through the material to detect changes within the substrate. Materials with areas of reduced thickness or lower density allows more radiation to pass through. The radiation, which passes through the material, forms a shadow image on a photographic film. Similar to UT, this method is able to detect flaws in metals and non-metals materials. Computed Tomography (CT) scanning is a sub-type of radiographic testing that compiles a series of X-ray images to produce a 3D image of a scanned part. ASTM E1441, E1742, E2033, E2445, E2698, E2736, E2737 and E3375 provide additional details in performing this method.

A. ADVANTAGES

RT is useful for nearly all substrates and is able to detect flaws from the surface to deep within the material to be inspected. RT is excellent at detecting discrete defects such as porosity and cracks and internal flaws in many different materials and produces a permanent image, or permanent 3D model in the case of a CT scan, as a result of the examination. The entire 3D part (or subset of the part) can be analyzed in a single inspection, producing results such as overall porosity within a part.

B. DISADVANTAGES

RT method usually needs gradients of density to create a detectable change in the test material. RT usually needs access to both sides of the area to be inspected. RT requires a large exclusion area or "vault" due to the hazardous nature of ionizing radiation. While handheld radiography units exist, the system is generally not field-portable due to the safety and power requirements, and part sizes are limited to the size of the "vault," generally up to 1 - 2' in any dimension. While RT is excellent at detecting discrete defects such as porosity and cracks, it is less reliable at for delaminations or subtle changes in density. When cracks are perpendicular to the direction of energy propagation, there is less contrast in the image and cracks are harder to detect.

C. APPROACHES FOR COLD SPRAY APPLICATIONS

Pre-Cold Spray: RT could be used to inspect for flaws at the surface/near-surface and deep within the test material.

Post-Cold Spray: RT could also be utilized to inspect for flaws at the surface/near-surface and deep within the cold spray deposit.

4.5.7 Thermography. Thermographic testing detects defects by imaging changes in the thermal diffusivity of a material and can be measured under passive or active conditions. The passive thermography method examines a component of interest using an infrared (IR) camera and differences in the thermal diffusive nature of the material are resolved. For active thermographic testing, a pulse of heat is deposited on the surface of the material, and the flow of heat is recorded using an IR camera as the heat diffuses into the material. When a flaw is present in the sample, it will affect the temperature distribution in the vicinity of the flaw and provide a contrast in the IR compared to the surrounding material. Thermography can detect pores, linear indications, delamination, and overall changes to a material's thermal diffusivity. However, this method is

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limited by the speed and resolution of the IR camera that is used. ASTM E2582 provides additional details in performing this method.

A. ADVANTAGES

It is a non-contact, full-field, single-side imaging method. Depending on the material's thermal diffusivity, thermography can detect both surface and subsurface defects.

B. DISADVANTAGES

Thermography has limited applicability with low thermal diffusivity or high-reflectivity materials. As with x-rays, the detection of linear indications is a function of how the thermal field is flowing with respect to the orientation of the linear feature, such as a crack. For surface breaking cracks, specialized thermography methods, such as line-scan thermography, have better capability that passive or flash thermography.

C. APPROACHES FOR COLD SPRAY APPLICATIONS

Pre-Cold Spray: Thermography testing could be utilized to inspect the surface of the substrate for cracks, pores, and other deficiencies in the bulk material.

Post-Cold Spray: Thermography testing could be utilized to inspect the entire cold spray deposit for cracks, pores, and discontinuities. It may be able to detect delamination at the substrate-cold spray surface.

5. DETAILED GUIDANCE

5.1 General. For all cold spray applications, MIL-STD-3021 may be used to ensure that operations meet an intended application's prescribed requirements and that those operations are repeatable and consistent. Additional qualification requirements may be required, such as those prescribed in MIL-STD-2035A, based on each specific application.

5.2 Cold Spray Structural and Non-Structural Applications. The effectiveness of cold spray technology is application dependent. MIL-STD-3021 defines cold spray applications as either structural or non-structural. Structural applications are where the cold spray deposit is a significant load-bearing member or where failure or disbonds could cause significant damage or reduction in service life. Non-structural applications are where the cold spray deposit is not a significant load-bearing member and where failure or disbonds do not cause significant damage or reduction in service life. Cold spray is most commonly used for applications where structural strength and service life of the repair are not essential (e.g., surface wear and dimensional restoration repairs). Additional information on non-structural and structural applications is available in Air Vehicle Subsystems Bulletin EN-ASTB-14-001, Structures Bulletin EZ-SB-13-001, and T00011542-GIB-010 for NAVSEA applications.

5.3 Cold Spray Approval. Cold spray technology approval is application dependent. MIL-STD-3021 provides cold spray approval guidelines for structural and non-structural applications. For nonstructural NAVSEA applications refer to T00011542-GIB-010. For structural cold spray applications and safety critical or foreign object damage critical applications, the cognizant engineering authority (CEA) is the member of the procuring activity that provides contract or purchase agreement technical input. The CEA will develop a qualification plan that follows the

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recommendations specified in Appendix A of MIL-STD-3021 and is specified in any applicable contract or purchase order. This should include but is not limited to MIL-DTL-32495A, which describes and lists powder properties that are required to identify and distinguish it from other powders.

5.4 Cold Spray Powders. Per MIL-DTL-32495, NDT methods may be specified by the contract and applicable CEA. For reference and later use in a production quality control program, these methods include but are not limited to electrical conductivity testing of the cold spray deposit per ASTM E1004. The CEA assumes all responsibility for assuring that all required testing and subsequent evaluation of any powders submitted to the preparing activity for listing under this specification are completed and pass all requirements specified herein and meet any additional performance criteria specific to their intended application.

5.5 Cold Spray Nondestructive Testing Approval. MIL-STD-3021 requires the CEA to perform NDT to detect relevant flaws of concern and its use should be specified in the contract or purchase order.

5.6 Cold Spray Processes, Quality Control, and Nondestructive Testing Requirements. T00011542-GIB-010 authorizes and establishes the requirements and procedures for application of cold spray within the Navy and Marine Corps. Requirements for conducting NDT, which are used in determining the presence of surface or internal flaws in metals, are contained in Naval Sea Systems Command Technical Publication T9074-AS-GIB-010/271. It also covers minimum requirements necessary to qualify NDT personnel, procedures, and equipment.

5.7 Cold Spray Airworthiness Approval. The application of cold spray is a modification to an air system, thus possibly affecting airworthiness. Airworthiness is the property of an air system configuration to safely attain, sustain, and complete flight in accordance with approved usage limits. An air system is an air vehicle, plus the training and support systems for the air vehicle (e.g., communications, control, ground/surface/control station, launch and recovery, and support elements), and any weapons to be employed on the air vehicle. Cited within multiple bulletins (see 2.2.2), airworthiness provides DoD personnel and contractors the appropriate level of safety of flight and risk management adapted to DoD-unique mission requirements.

5.8 Cold Spray Quality Assurance (QA). Hardness and adhesion of sprayed deposits, such as cold spray, are an essential part of assessing the quality of repair. NDT methods for assessing the adhesion quality of sprayed deposits have had limited success to date, however, progress continues to be made. The ultimate goal for NDT of CS applications is to reduce the need for destructive testing.

5.9 Summary of Cold Spray NDT Challenges. At present, current NDT methods do not reliably measure cold spray coating features such as cracking, porosity, oxide levels, coating disbond, or adhesion. Crack detection by dye-penetrate methods often fails to detect isolated defects against the general presence of porosity. Radiography, eddy current and ultrasonic inspections are of little use because of the heterogeneous nature of sprayed coatings and the small dimensions of features and defects. As a consequence to the limited success of NDT currently, greater emphasis is given to destructive methods such as adhesion tests and metallography. Destructive testing on a small percentage of repairs is acceptable where large numbers of similar articles are being processed. However, many thermal spraying jobs involve coating small numbers of components, and in

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reclamation work, it is typical for only one part to be repaired. In these circumstances it is necessary to coat a test coupon. It is essential to ensure that the coating on the test coupon is similar to that on the component. This requires not only consistency in spraying conditions but also an appreciation of the effects of, for example, differences in component section thickness and cooling rate on deposit structure and properties.

6. NOTES

6.1 Intended use. This handbook provides Department of Defense (DoD) recommended guidance to address nondestructive testing (NDT) techniques and quality assurance (QA) processes in cold spray applications. It covers NDT techniques including visual, liquid penetrant, magnetic particle, eddy current, ultrasonic, radiographic testing, and thermography. The effectiveness of each NDT technique is discussed within this document.

6.2 Cold Spray Categorization. The term "cold spray" refers to the relatively low process temperature involved, which is typically much lower than the spray material's melting point. While categorized as a "thermal spray" process, cold spray is not a true thermal spray process because it does not use thermal energy as the primary energy source to melt materials. Instead, cold spray uses kinetic energy rather than heat to promote adhesion to the substrate.

6.3 Additive Manufacturing or Repair. Cold spray is sometimes referred to as an "additive manufacturing" or "additive repair" process. For additive manufacturing or repair, material is typically added in a precision manner under automated control (often guided by a digital model of the component being fabricated) and a minimum amount of finishing work is required to complete the fabricated part. In cold spray, material is typically deposited and then the part is machined to final dimensions. Therefore, cold spray is best referred to as a "material addition" process.

6.4 Key Words.

Additive	Ferrous	Nondestructive Inspection
Air system	Ferromagnetic	Non-ferrous
Airworthiness	Flaw	Non-metal
Cognizant Engineering Authority	Indication	Non-structural
Defect	Metal	Repair
Deposition	NDI	Quality Assurance
Discontinuity	NDT	Structural

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APPENDIX A

A.1 SCOPE

A.1.1 Scope. This Appendix provides an illustration of the Cold Spray Process which is a restorative additive manufacturing technology to build up material on parts. It is used for metallic structural repair of similar or dissimilar materials and is ideal for repairs since it does not affect substrate heat treatment.

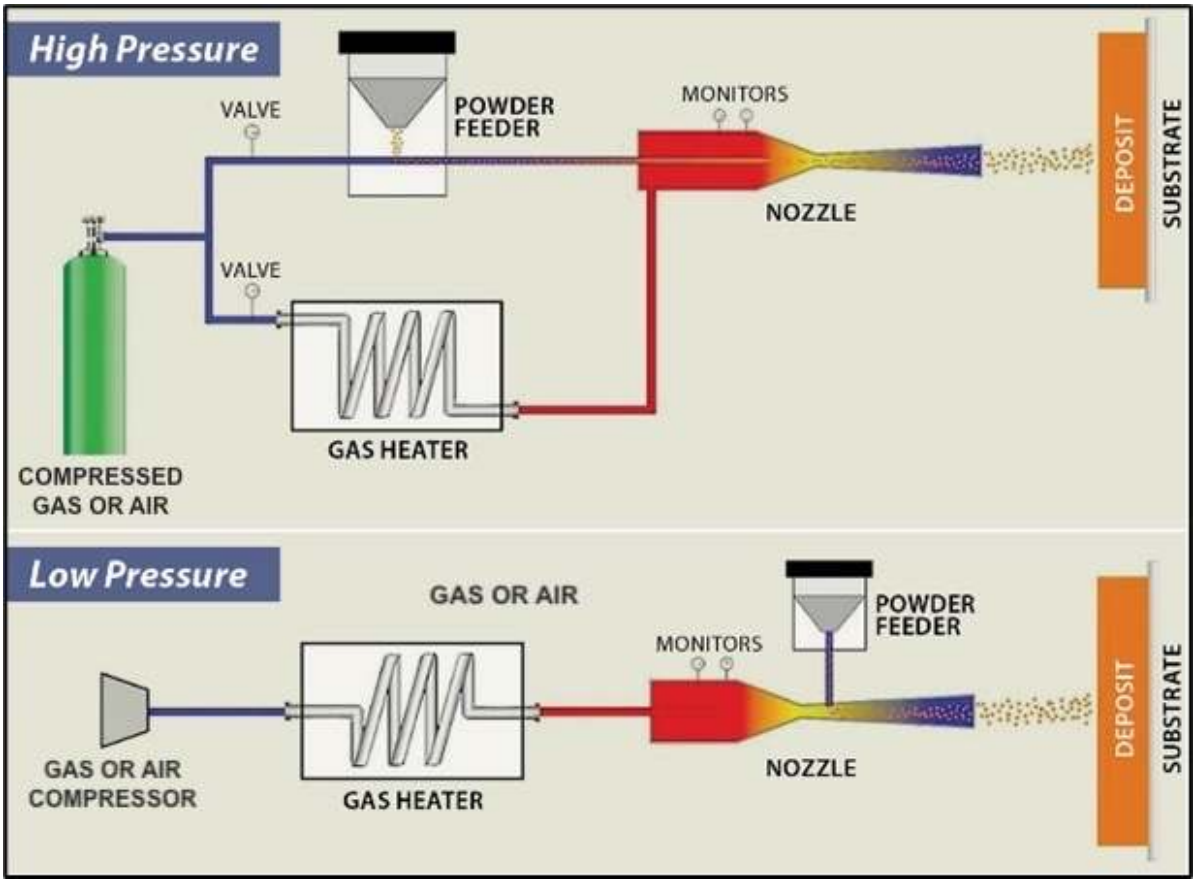


FIGURE A-1. Cold Spray Process Illustration.

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APPENDIX B

B.1 SCOPE

B.1.1 Scope. This appendix tabulates the various types of Nondestructive Testing methods, their applicability to Metallic and Non-Metallic materials (table B-I), and summarizes their respective required resources, utility, versatility, and limitations (table B-II).

TABLE B-I. Nondestructive Testing Material Types.

NDT Method	Metallic	Non-Metallic
Visual	✓	✓
Liquid Penetrant	✓	✓
Magnetic Particle	✓[1/]	×
Eddy Current	✓	×
Ultrasonic	✓	✓
Radiography	✓	✓
Thermography	✓	✓

[1/] Ferromagnetic Metal

TABLE B-II. Nondestructive Testing Methods Summary.

NDT Method	Resources Required	Utility of Method	Versatility of Method	Limitations of Method
Visual	Low	Low	High	Surface Inspection
Liquid Penetrant	Low	Low	Mid	Surface Inspection/ Darkened Area (FPI)
Magnetic Particle	Low/ Mid	Mid	Low	Ferromagnetic Only
Eddy Current	Mid	Mid	Mid	Near Surface/ Representative Standard
Ultrasonic	Mid	High	Mid	Surface Condition
Radiography	High	Mid	Low	Access Limitations
Thermography	Mid	Mid	Mid	Near Surface/ Low Emissivity Difficult

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CONCLUDING MATERIAL

Custodians:
Army - MR
Navy - AS
Air Force - 20

Review activities:
Army - AV, CR, PT
Navy - EC, MC, SH
Air Force - 10, 11, 16, 19
DLA - DH, GS8

Preparing Activity:
Army - MR

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NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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